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ANOMALY DETECTION IN OIL-PRODUCING WELLS USING LSTM AUTOENCODER

K. C. Schmöckel
G. F. Peixer
J. A. Lozano Cadena
POLO – Research Laboratories for Emerging Technologies in Cooling and Thermophysics, Department of Mechanical Engineering, Federal University of Santa Catarina (UFSC), Florianópolis, SC, 88040-900, Brazil.
kevin.costa@polo.ufsc.br
guilherme.peixer@polo.ufsc.br
jaime@polo.ufsc.br

J. R. Barbosa, Jr.
Craft & Hawkins Department of Petroleum Engineering, Louisiana State University, Baton Rouge, LA 70803, United States
jbarbosa@lsu.edu

Abstract. *Oil-producing wells need careful management and can significantly benefit from advanced technologies and techniques. These are especially important for preventing undesirable events that could lead to environmental accidents, higher maintenance costs, and reduced production. Artificial intelligence (AI) and machine learning (ML) techniques have proven effective for detecting anomalies in these situations. Petrobras developed the 3W dataset, which is a public collection of multivariate time series data. This dataset is valuable for creating and improving methods to detect anomalies. It includes real data from oil-producing wells, simulated data, and data drawn by hand by experts in the field. Many studies have used this realistic dataset to develop advanced methods for identifying different types of undesirable events. A common approach in these studies involves statistical methods, such as Isolation Forest, Local Outlier Factor, and One-class Support Vector Machine. In this particular study, we employed an Autoencoder model with a Long Short-Term Memory (LSTM) architecture to detect anomalies. We exclusively used the real data from the 3W 2.0.0 dataset, which is its latest version. The Autoencoder works by compressing the data and then reconstructing it. Anomalies are identified and categorized by comparing the reconstructed data to the original data, looking for significant differences or 'deviations' that exceed certain set limits or 'thresholds.' The LSTM architecture is known for its strong performance in detecting anomalies across various datasets. The F1 score, a widely recognized evaluation metric, was used to assess the model's performance.*

Keywords: *Anomaly detection, Oil-producing wells, Autoencoder*

1. INTRODUCTION

Oil derivatives are currently present in many aspects of our lives and impact various sectors in the contemporary world (Florescu *et al*., 2019). Oil plays a vital role in many countries' social and industrial development, serving as a primary source of energy and a cornerstone of the modern economy (Shi et al., 2021). This makes the methods we use to find, produce, and refine it incredibly important. Since its discovery in the 19th century, the oil demand has steadily increased, highlighting the ongoing need to develop new techniques, technologies, and scientific understanding in this field. (Elanchezhiyan *et al*., 2016).

Oil-producing wells are located worldwide, and their exploration without proper measures can lead to significant environmental problems. Many of these problems can be prevented by following established security and safety standards. However, other issues are not so easily resolved, which highlights the need to understand patterns in oil-producing wells. Predicting undesirable events during production is a crucial defense against potential environmental damage and financial losses. (Mahmood *et al*., 2023).

The rise of this necessity has led many scholars and academic professionals to look for methods to predict anomalies in oil wells (Jambol *et al*., 2024). Machine learning techniques play a fundamental role in accomplishing such tasks and allow proactive maintenance and mitigation strategies (Lin *et al*., 2022). Algorithms have been developed and up to date have been helpful and presented good results, not only for oil and gas applications, but also for other types of anomaly detection, such as human health. Anomaly detection techniques have advanced incredibly fast in the past few years, and many of them are still being created. Among those that have received special acknowledgment are Isolation Forest, Local Outlier Factor, and One-class Support Vector Machine (Júnior *et al*., 2020). By analyzing the difference between predicted and real values, anomalies can be detected, and the appropriate decisions can be taken (Ha, 2022).

Developing an anomaly detection method is not trivial, and a fundamental support is a reliable dataset on which the models will be trained. In 2019, Petrobras (Petróleo Brasileiro S.A.) published the 3W dataset, which includes real, simulated, and hand-drawn data taken from real oil wells, containing values of temperature, pressure, and mass flow in different parts of production (Vargas *et al*., 2019). Recently, they have compiled and published an update, bringing more

series from different wells, turning the already important dataset into an even more robust one. The purpose of making 3W public is to support researchers in developing state-of-the-art techniques to detect anomalies in oil wells. Implementing machine-learning techniques can benefit oil and gas companies, like Petrobrás, to detect anomalies and reduce financial and environmental losses (Canizo *et al*., 2019). Since this publication, many studies have been conducted in universities around the world and especially in Brazil, looking to improve the already existing models (Nascimento *et al*., 2021).

At first, many statistical architectures were implemented to locate undesirable events using the 3W dataset. Some of them have presented efficiency, although these traditional methodologies are being outperformed by deep learning algorithms. Most recently, researchers have focused on the newest techniques (Fernandes Jr *et al*., 2024). Many studies have proved the prowess of LSTM (Long Short-Term Memory) compared to statistical algorithms when it comes to anomaly detection (Machado *et al*., 2022). For that, this study proposes to test an LSTM Autoencoder detecting anomalies in the most recent update of 3W and evaluating its results, considering that this same architecture was tested before in the first version of the dataset in Villamil *et al.* (2024).

2. 3W 2.0 DATASET

2.1 OFFSHORE NATURALLY FLOWING OIL WELLS

In July of 2019, Petrobras released the 3W dataset, which includes a compilation of time series data that correlates features present in an oil well to abnormal events on it. The present wells in the dataset are offshore naturally flowing wells, and they are characterized by having a sufficient reservoir pressure to produce hydrocarbons at a commercial rate without the use of any extra energy. Each oil well has a set of transducers, sensors, and mechanical, pneumatic, or hydraulic systems that could be installed on the seabed, the surface, or in the well itself to extract petroleum and natural gas (Guo, 2011). Naturally flowing oil wells usually have less equipment installed due to their capacity to produce oil naturally. Fig. 1 presents a simplified schematic of a typical offshore installation (Vargas *et al*., 2019). As seen in Fig.1, hydrocarbons are transported via a production line, and valves and sensors are operated via an electro-hydraulic umbilical. During normal operation, devices like pressure and temperature transducers register and save fluid features, generating multivariate time series.

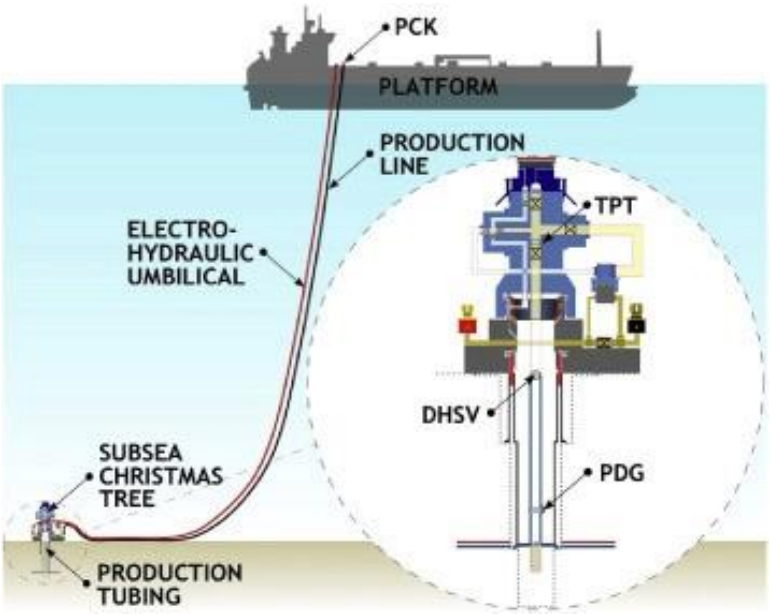

Figure 1. Schematic representation of a naturally flowing offshore oil well, from Vargas *et al*. (2019)

Through continuous monitoring of variables taken from sensors and transducers, it is possible to know if the operation of the well is under normal conditions or if an anomaly is happening. The most common monitored variables in Petrobras offshore naturally flowing oil wells include pressure and temperature at the TPT and the Permanent Downhole Gauge (PDG), respectively, and their positions can be seen in Fig. 1. Flow of gas lift, pressure, and temperature at different points are also important and monitored.

2.2 THE DATASET

The 3W dataset compiles recorded values of many different features throughout time in an oil well, characterizing a dataset of multivariate time series data. Every feature is a variable, and all of them are collected, creating a series (Vargas *et al*., 2019). Not only are real data taken from real wells present in the dataset, but hand-drawn series by experts and simulated data are also covered in the dataset. By having not only real events, different types and possibilities of anomalies in the dataset, and helps it to cover a bigger range of possible cases.

When released, 3W was the first publicly available dataset containing real abnormal events in oil wells. It had multivariate time series data taken from 18 different wells. In July of 2024, when update 2.0.0 came out, data from 42 different wells were available. The most recent update brought 20 new variables, totaling 27, labeling gaps present in the first version were eliminated, and a different type of anomaly was added after the addition of real and simulated series containing the new class 9. Several instances of different already present anomaly types were added, normal periods in many real instances with unwanted events were extended, corrections were made to conversions between measurement units in several real instances, and a few more changes were made to the already robust dataset.

The dataset has 9 different types of anomalies present on it and they can be seen in Tab. 1. The Petrobras project focused on offshore naturally flowing oil wells that are in a normal state, therefore all the abnormal events present on the 3W dataset began from a normal operation condition and series that don't have an abnormal state receive the class number 0. More detailed explanations about all of them can be found at Vargas *et al*. (2019)

Table 1. Summary of event types in the 3W dataset

Class	Type of event
0	Normal Operation
1	Abrupt Increase of BSW
2	Spurious Closure of DHSV
3	Severe Slugging
4	Flow Instability
5	Rapid Productivity Loss
6	Quick Restriction in PCK
7	Sealing in PCK
8	Hydrate in Production Line
9	Hydrate in the Service Line

Among the features brought by the dataset, there is the column of state, which indicates if the well is under normal or abnormal operation. Considering that every series present in the dataset has begun as a normal state, every series that has an anomaly detected has three different labeled periods, which are normal, faulty transient, and faulty steady state. In the faulty transient state, the dynamics resulting from an undesirable event are still ongoing. Fig. 2 and Fig. 3 present two series taken from 3W 2.0.0 that started under normal conditions and then became abnormal with a problem of hydrate in the service line, which is class 9. In both cases, there are 5 features presented, and different periods have different colors, blue for normal, yellow for faulty transient condition, and red for the anomalous state.

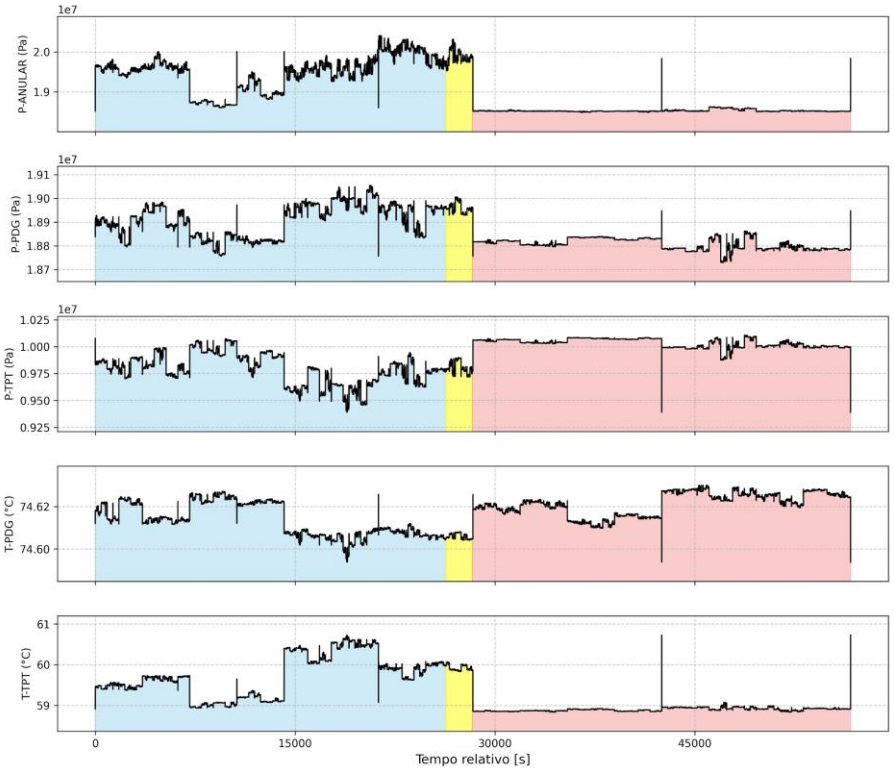

Figure 2. Real instance of class 9 taken from Well 42 on 2014 December 18th

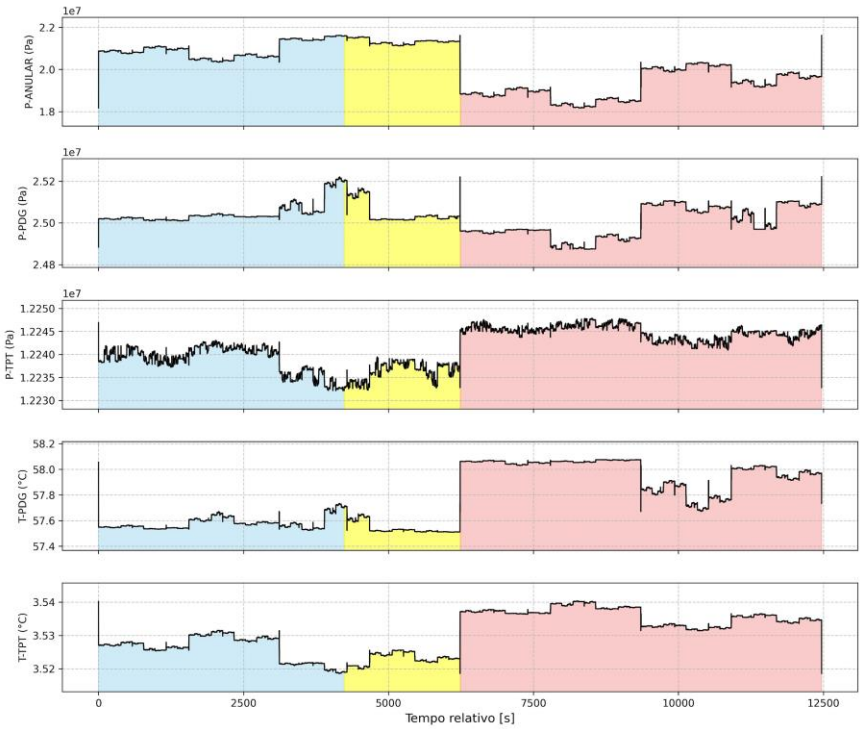

Figure 3. Real instance of class 9 taken from Well 16 on 2015 April 18th

It is important to note that different features influence the state of an oil well; however, not all of them have the same level of influence, nor are they all positive or negative. It turns out to be a necessity to learn the real correlation between an input variable and an undesirable event. In Fig. 4, a Correlation Matrix is presented, developed using different series in the 3W dataset, and each column brings the Pearson correlation coefficient of a feature with the corresponding anomaly. It can be observed that temperature at TPT has more impact on anomalies of types 3 and 4 than on any other, and pressure at PDG has more influence on anomaly class 8 than on any other anomaly. However, every feature present on the matrix has an impact on the present anomalies, so it strengthens the importance of considering all variables when developing an

anomaly detection model and the importance of making an algorithm capable of understanding and learning the complexity of all abnormal events under evaluation.

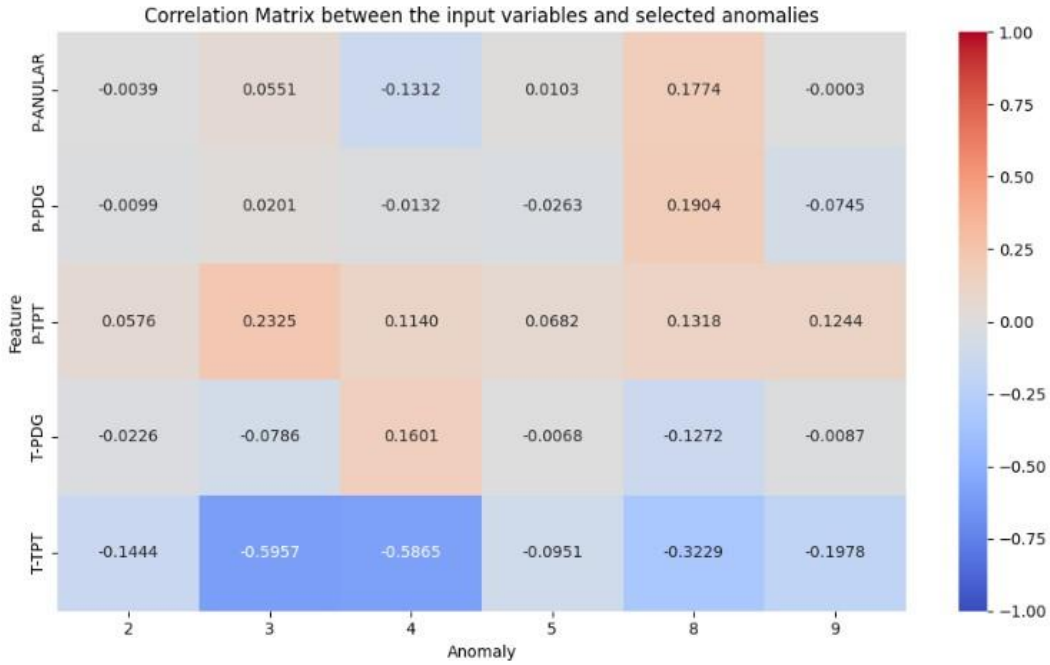

Figure 4. Correlation Matrix between the input variables and selected anomalies.

3. MODEL DEVELOPMENT

This work focused on the evaluation of an LSTM Autoencoder to detect anomalies in series taken from 3W 2.0.0. The series was split into segments of 180 seconds and divided into positive or negative. If the segment had any anomalous state present on it, then it is considered positive; unless it happens, the segment is negative. Then, some rules were adopted to improve the algorithm results.

The first rule was that only series with an undesirable event type would be used, and every series had to have a normal state for at least 30 minutes to be used. Another rule was that only negative segments would be used during training, and for testing, both positive and negative segments would be used. From the whole dataset, 80% of the segments were used for training, 10 % for validation, and 10% for testing; the number of samples of each anomaly was correspondingly divided into each category of use.

3.1 DATA PREPROCESSING

Data preprocessing is an essential step to be taken before utilizing a dataset. Especially in the field of anomaly detection. It is important to determine how to manage missing and frozen variables, and what instances should be discarded or not. As previously mentioned, the 3W dataset has real, hand-drawn, and simulated instances, and it was decided to only use real data in the present work, so hand-drawn and simulated cases were removed from the evaluation.

Raw data taken from the dataset was divided into 482 series, and the division occurred by gathering data from the same well together and uniting subsequent timesteps. When gaps were found, another series was created. This way every series has a well which it belongs to and all the timesteps are connected without breaks of seconds, and the series are named after the well, the date, and the second it starts. A problem present in 3W is the amount of missing values. These values were accounted for in each series and for every feature. The number of features in every series is 27, but when the percentage of missing values in one of these features exceeded 10%, the data series was disregarded they were disregarded.

Of the 27 features, 9 had over 50% missing values considering every instance, and so, it was decided to discard these features from all the series. A more accurate decision of which features preserve for modeling was made considering the graphic of Fig. 5. On the X axis are represented the features present in the series and on the Y axis are represented the amount of removed series to have those features present on the evaluated dataset considering that every feature has to have less than 10% of missing values in every instance. For example, to have ESTADO-M1 and all the other features of the graphic included during model evaluation, over 400 series would have to be discarded.

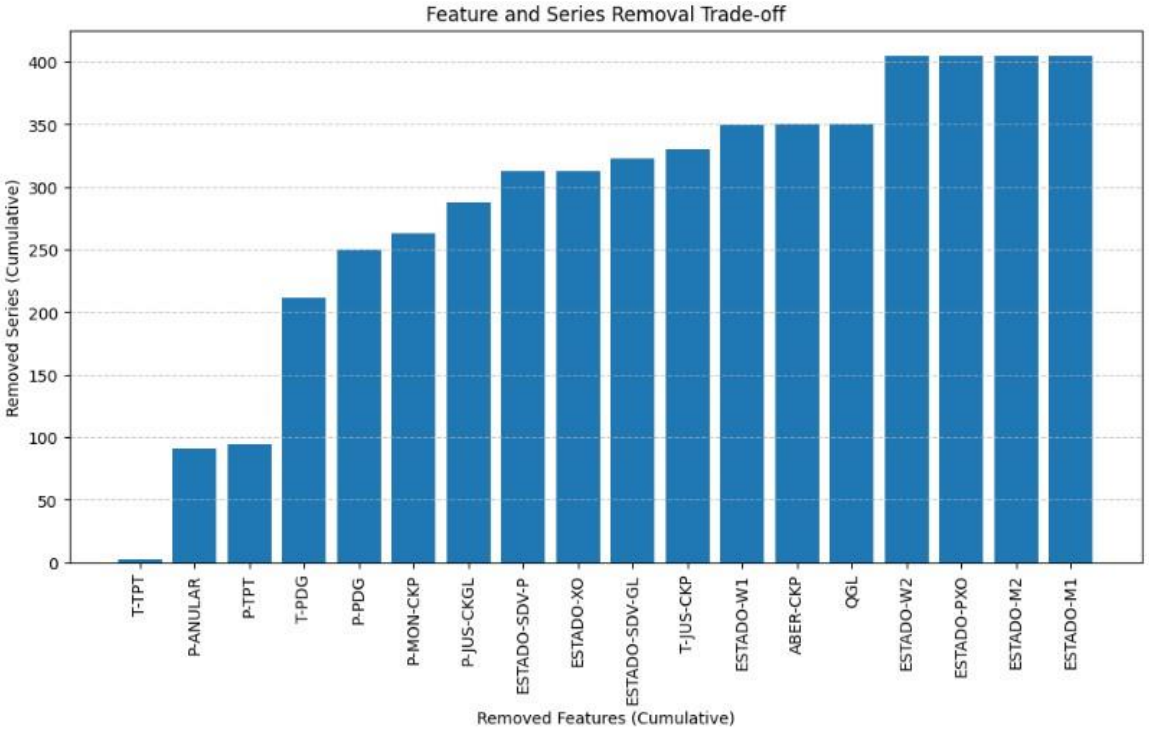

Figure 5. Graphics representing the features necessary to preserve or remove for having a certain amount of series with less than 10% of missing values for every feature in it.

The optimum case would be to have all the 482 series with all the features, but we have to do a trade-off between a good number of series and a good number of features. It was chosen to test the model with 5 features, which would be P-PDG, T-PDG, P-TPT, P-ANULAR, and T-TPT, and with these features having less than 10% missing values in every series, we can use 232 series during analysis. It is important to know that every series has a minimum of 10 negative segments, 30 minutes of normal operation as previously mentioned, and 4 positive segments. Considering that, it was not necessary to do an oversampling with the data or the use of sliding windows, because the amount of data, even with everything that was discarded, is robust enough to train and test the model. The missing values of the series included during analysis were replaced by the median of close timesteps in the same column, and all the data were normalized by the mean and standard deviation of the data in the training set.

3.2 MODEL SELECTION

It is known that the LSTM autoencoders are being employed for many different applications and outperform traditional statistical methodologies for anomaly detection (Machado *et al*., 2022). It can handle sequential data and to capture temporal dependencies, which is why it is considered one of the most promising deep learning architectures for anomaly detection in time series (Niu *et al*., 2020), and it was selected for use.

LSTM Autoencoders encode time series into a latent representation and then decode it, which means reconstructing it to the initial configuration from the latent space. During the training phase, the LSTM Autoencoder learns normal data patterns and improves its capacity to identify outliers in a segment. Thus, good training will empower the ability to distinguish between normal and anomalous states. The encoder and decoder are composed of 2 layers, each with 32 LSTM units in the present work. Both layers use the hyperbolic tangent activation function. To prevent overfitting, early stopping is used, and the Adam optimizer (Kingma, 2014) is also used for training the model, with a learning rate of 0,0001. LSTM architecture offers many advantages when compared to other methodologies to accomplish outlier identification tasks, but it is not perfect. Some shortcomings could be loss of information in the LSTM network (Bashar and Nayak, 2023), the complexity of training the encoder and decoder (Niu et al., 2020), and the time it requires.

3.3 PERFORMANCE METRICS

The model was evaluated by the F1 score, a traditional performance metric that allows comparing these work results with other ones encountered in the literature. It can be calculated by:

$$F1 = 2\frac{Precision.Recall}{Precision+Recall} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \tag{2}$$

$$Recall = \frac{TP}{TP+FN} \tag{3}$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives (Elmrabit *et al*., 2020).

4. RESULTS

The results are presented according to the metrics of F1 score and the confusion matrix present in Fig. 6. The confusion matrix correlates the actual states of the oil wells and the predicted states by the LSTM Autoencoder, showing the rate of correlation. The F1-score obtained was 0,9715.

Considering the results, the model presents a very good performance and is in line with the results found in the literature. Analysing the confusion matrix, we can see that the model can flag better when a segment is anomalous than when it is normal, and this is positive, because when applied to a real oil well, the model would always identify the anomalies and notify users.

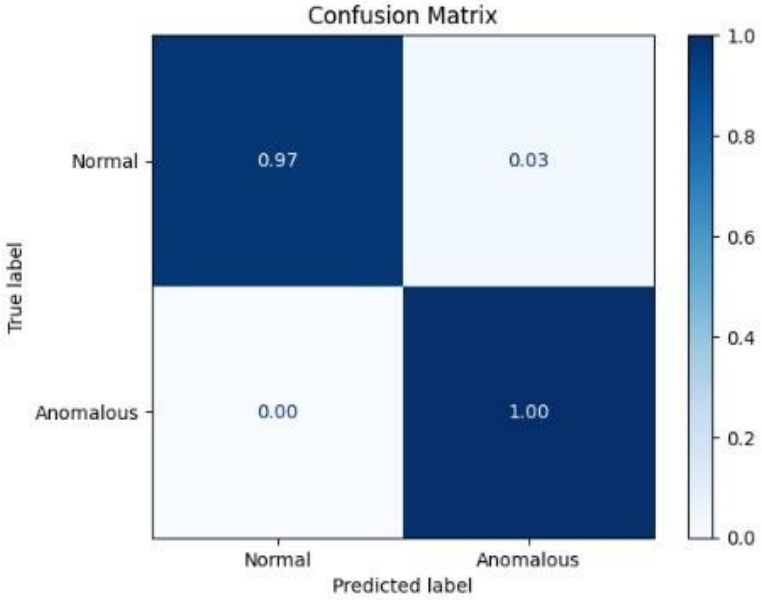

Figure 6. Confusion Matrix of the developed model on the test set.

5. CONCLUSIONS

This study successfully applied an LSTM Autoencoder to detect anomalies in naturally flowing oil wells. We evaluated its performance using the 3W dataset, a comprehensive collection of multivariate time series data from real, simulated, and expert-drawn oil wells. Petrobras created this publicly available dataset to foster the development of cutting-edge techniques for preventing environmental disasters and economic losses. After careful data preprocessing, the LSTM Autoencoder was trained and tested, yielding excellent results and demonstrating its effectiveness in identifying and classifying undesirable events.

While the model proved highly effective at determining when an anomalous state occurred, it occasionally indicated an anomaly even when conditions were normal, leading to unnecessary alerts. This suggests there's still room for performance improvement. The F1-score of 0.9715 reflects a strong balance between precision and recall, but it also highlights this concern regarding false positives. Despite this challenge, the LSTM Autoencoder proved to be a valuable tool for anomaly detection and can be effectively implemented with the 3W dataset. Its capacity to handle multivariate time series data and its customizable features—such as the number of layers, cells, regularization rates, and other parameters—make it a promising direction for future research.

Errors in oil production can lead to severe environmental issues, increased maintenance costs, and production losses. Therefore, it's crucial to optimize these models for practical use in real wells. Future research could focus on optimizing the current model, testing it with different preprocessing techniques, and even incorporating simulated and hand-drawn data. Combining the LSTM Autoencoder with other architectures and machine learning methods, like transformer networks, also presents exciting possibilities. The field offers vast opportunities for exploring new techniques, and ongoing engineering developments can help solve current challenges in oil production.

6. ACKNOWLEDGEMENTS

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